SKELGYM BIOMECHANICAL TOPOLOGY: COMPLETE 24 KINEMATIC ANGLES

Unified Representation: 14 Limb Elevations ($\theta_{1\dots14}$, Blue) and 10 Joint Articulations ($\psi_{15\dots24}$, Orange) on MediaPipe 3D Pose

14 LIMB SEGMENT ELEVATIONS (θ)

$\boldsymbol{\theta} = \arctan2(\boldsymbol{\Delta y}, \sqrt{\boldsymbol{\Delta x^2 + \Delta z^2}}) \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

Vertical inclination w.r.t ground X--Z (0° horiz, $\pm 90^\circ$ vert)

■ UPPER LIMBS & FOREARMS

θ_1 : Left Upper Arm Elevation
$\vec{v} = \mathbf{p}_{13} - \mathbf{p}_{11}$ (Shoulder → Elbow)

θ_2 : Right Upper Arm Elevation
$\vec{v} = \mathbf{p}_{14} - \mathbf{p}_{12}$ (Shoulder → Elbow)

θ_3 : Left Forearm Elevation
$\vec{v} = \mathbf{p}_{15} - \mathbf{p}_{13}$ (Elbow → Wrist)

θ_4 : Right Forearm Elevation
$\vec{v} = \mathbf{p}_{16} - \mathbf{p}_{14}$ (Elbow → Wrist)

■ LOWER LIMBS & SHANKS

θ_5 : Left Thigh Elevation
$\vec{v} = \mathbf{p}_{25} - \mathbf{p}_{23}$ (Hip → Knee)

θ_6 : Right Thigh Elevation
$\vec{v} = \mathbf{p}_{26} - \mathbf{p}_{24}$ (Hip → Knee)

θ_7 : Left Shank / Shin Elevation
$\vec{v} = \mathbf{p}_{27} - \mathbf{p}_{25}$ (Knee → Ankle)

θ_8 : Right Shank / Shin Elevation
$\vec{v} = \mathbf{p}_{28} - \mathbf{p}_{26}$ (Knee → Ankle)

■ AXIAL TORSO & GIRDLES

θ_9 : Shoulder Girdle Clavicular Tilt
$\vec{v} = \mathbf{p}_{11} - \mathbf{p}_{12}$ (Clavicular bar tilt)

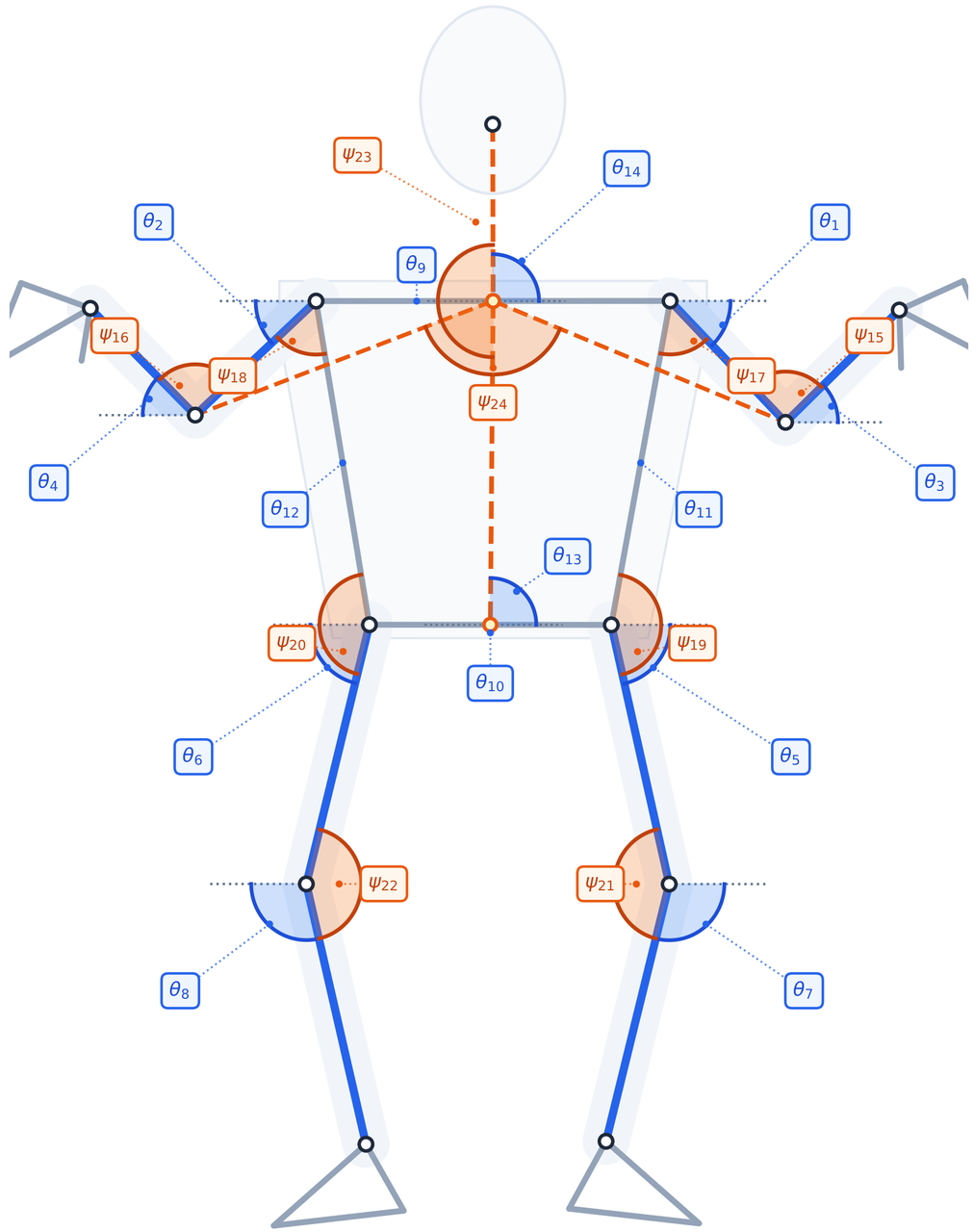
θ_{10} : Pelvic Girdle Hips Tilt
$\vec{v} = \mathbf{p}_{23} - \mathbf{p}_{24}$ (Pelvic bar tilt)

θ_{11} : Left Torso Flank Tilt
$\vec{v} = \mathbf{p}_{23} - \mathbf{p}_{11}$ (Shoulder → Hip)

θ_{12} : Right Torso Flank Tilt
$\vec{v} = \mathbf{p}_{24} - \mathbf{p}_{12}$ (Shoulder → Hip)

θ_{13} : Longitudinal Spine Axis Tilt
$\vec{v} = \mathbf{p}_{\text{mid_shoulder}} - \mathbf{p}_{\text{mid_hip}}$

θ_{14} : Cervical Neck/Head Axis Tilt
$\vec{v} = \mathbf{p}_{0} - \mathbf{p}_{\text{mid_shoulder}}$ (Shoulder → Nose)



10 JOINT ARTICULATIONS (ψ)

$\boldsymbol{\psi} = \arccos\left(\frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}\right) \in [0, \pi]$

Anatomical Triplet Inner Angles: Flexion, Extension, Posture

■ UPPER EXTREMITY FLEXION & EXTENSION

ψ_{15} : Left Elbow Flexion Angle
Triplet: $\angle(\mathbf{p}_{11}, \mathbf{p}_{13}, \mathbf{p}_{15})$ (Shoulder--Elbow--Wrist)

ψ_{16} : Right Elbow Flexion Angle
Triplet: $\angle(\mathbf{p}_{12}, \mathbf{p}_{14}, \mathbf{p}_{16})$ (Shoulder--Elbow--Wrist)

ψ_{17} : Left Glenohumeral Shoulder Angle
Triplet: $\angle(\mathbf{p}_{23}, \mathbf{p}_{11}, \mathbf{p}_{13})$ (Hip--Shoulder--Elbow)

ψ_{18} : Right Glenohumeral Shoulder Angle
Triplet: $\angle(\mathbf{p}_{24}, \mathbf{p}_{12}, \mathbf{p}_{14})$ (Hip--Shoulder--Elbow)

■ LOWER EXTREMITY FLEXION & EXTENSION

ψ_{19} : Left Acetabulofemoral Hip Hinge
Triplet: $\angle(\mathbf{p}_{11}, \mathbf{p}_{23}, \mathbf{p}_{25})$ (Shoulder--Hip--Knee)

ψ_{20} : Right Acetabulofemoral Hip Hinge
Triplet: $\angle(\mathbf{p}_{12}, \mathbf{p}_{24}, \mathbf{p}_{26})$ (Shoulder--Hip--Knee)

ψ_{21} : Left Tibiofemoral Knee Flexion
Triplet: $\angle(\mathbf{p}_{23}, \mathbf{p}_{25}, \mathbf{p}_{27})$ (Hip--Knee--Ankle)

ψ_{22} : Right Tibiofemoral Knee Flexion
Triplet: $\angle(\mathbf{p}_{24}, \mathbf{p}_{26}, \mathbf{p}_{28})$ (Hip--Knee--Ankle)

■ AXIAL & TRUNK KINEMATICS

ψ_{23} : Sagittal Torso Posture Angle
Triplet: $\angle(\mathbf{p}_{0}, \mathbf{p}_{\text{mid_shoulder}}, \mathbf{p}_{\text{mid_hip}})$
Tracks spinal erectness vs. trunk forward flexion

ψ_{24} : Coronal Bilateral Arm Splay Angle
Triplet: $\angle(\mathbf{p}_{13}, \mathbf{p}_{\text{mid_shoulder}}, \mathbf{p}_{14})$
Measures bilateral elbow excursion & lateral abduction

SkelGym Compound Representation : $\mathbf{I}_{\text{mix}} = \mathbf{I}_{\text{world_3d}}\,(39\text{-d}) \oplus \mathbf{I}_{\text{kinematic_24}}\,(24\text{-d}) \in \mathbb{R}^{63}$ • **Inherently invariant to horizontal camera yaw rotation** $R_y(\alpha) \in \mathrm{SO}(2)$